



Social Media Data – Twitter/X

Web Science
COMPSCI4077/COMPSCI5107/COMPSCI5078

Joemon M Jose

Overall info

announcements

Communication

- Through Teams Page
 - so please sign in -
 - [WEB SCIENCE - COMPSCI4077, COMPSCI5078 & COMPSCI5107 - 2024-25 | General | Microsoft Teams](#)
 - If you are not on Teams Channel
 - you will not get any information/emails
- Lecture materials
 - Moodle
 - <https://moodle.gla.ac.uk/course/view.php?id=44995>
 - Coursework & deadlines
 - Announcement on Week 2

Which module to enroll?

- COMPSCI4077 - Web Science H – For level H students
- COMPSCI5107 - Web Science for MSc
- COMPSCI5078 - Web Science M – For Level M students
- Please enrol for the right module
 - Otherwise affect your degree!!!

Please watch the Introduction lecture

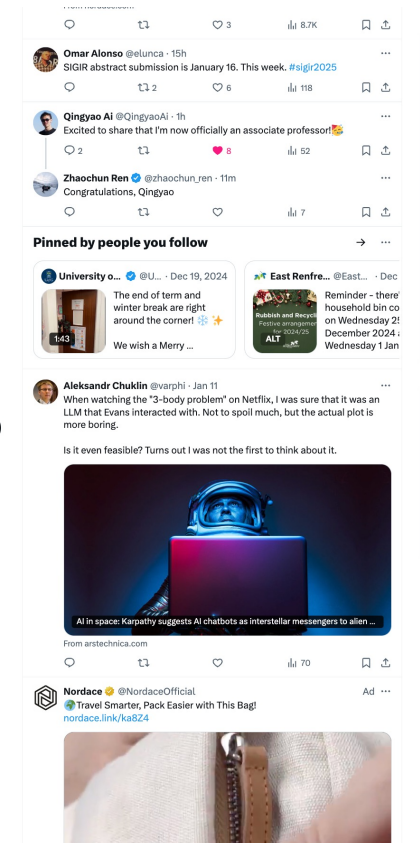
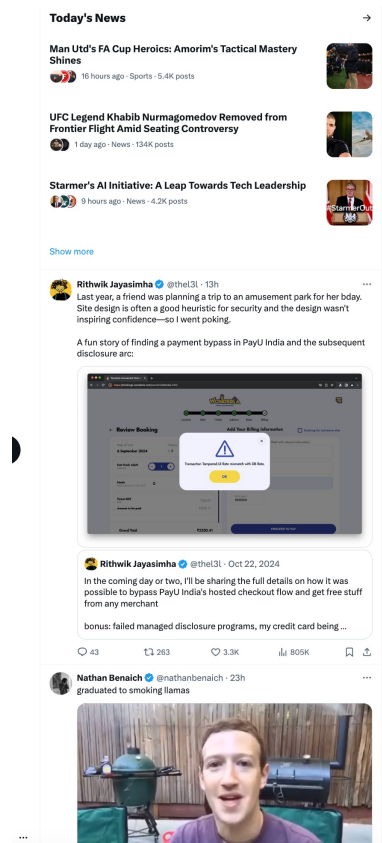
- If you have any questions, please write your questions at
 - https://glamys.sharepoint.com/personal/joemon_jose_glasgow_ac_uk/_layouts/15/doc.aspx?sourcedoc={9185079d-37bf-48dd-84db-aa34e6aeb910}&action=edit
 - I will answer them in the class
 - This will help the rest of the class as well
 - they will get an opportunity to clarify issues

PART I

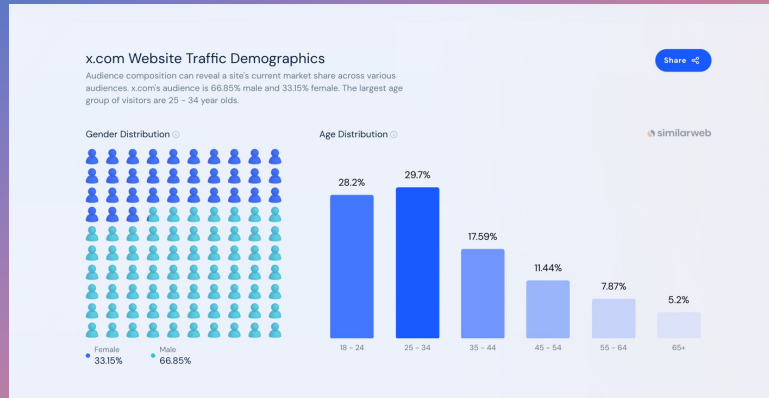
Background

Social media & making sense

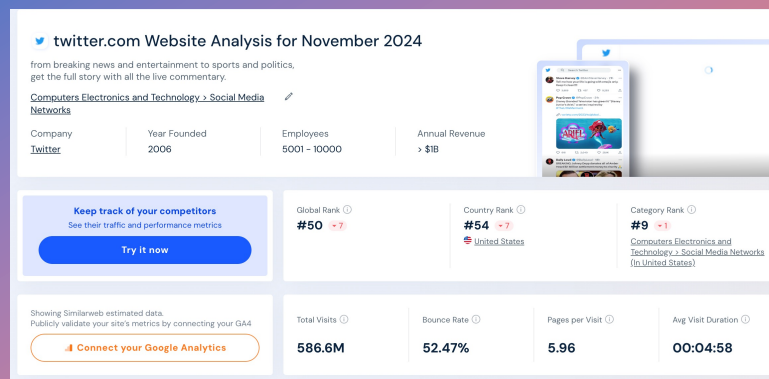
- Millions of people communicate their feelings and share information through social media
 - Facebook
 - Twitter
 - Instagram
 - SnapChat
 - Reddit
- Can we exploit such public communication to study and understand the underlying social phenomenon?
 - What do people think about COVID booster programs?
 - Normally social scientists' study this, through
 - Gallop poll - the company became known for its public opinion polls conducted worldwide.
- Could we do better and faster?

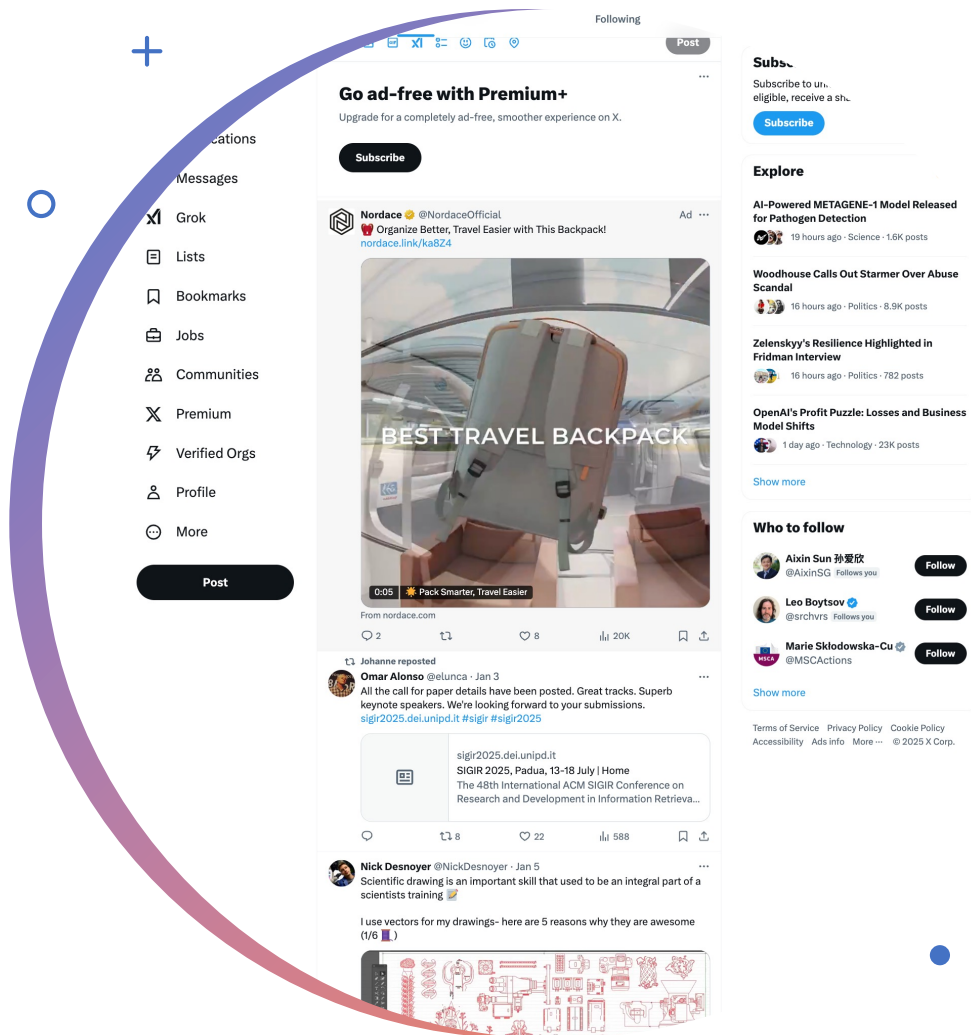


Twitter/X - Stats



- Ranked 5th most popular site in the world
 - <https://www.similarweb.com/website/x.com/#overview>
 - During the 2014 FIFA world cup final, 618, 725 tweets were sent in a single minute
- No of daily tweets
 - 4.5B visits per month;
 - As of early 2024, X had an estimated 429 million users worldwide.
 - Most users are under 35 years old, with 28.2% of users between 18 and 24 years old,
 - and 29.7% between 25 and 34 years old.
- **Advertising revenue**
 - In 2023, X's annual advertising revenue was \$2.98 billion.
- **Past Statistics**
 - 134 -139 million active users daily! ~42% of users
 - 92% of US Tweets come from 10% of Twitter users





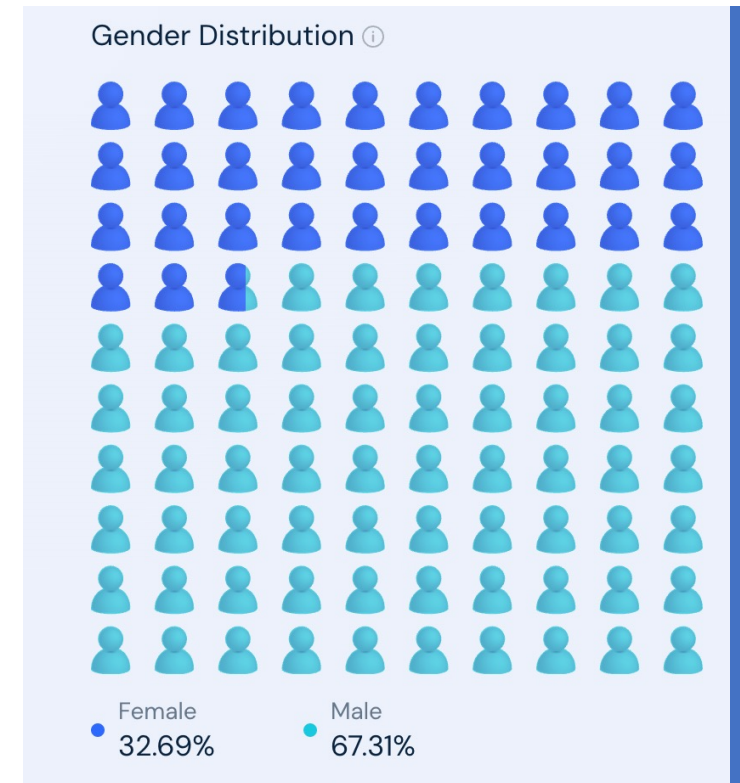
Importance of X platform!

- Vital communications platform in times of disasters
 - Hurricanes, earthquakes, political upheavals, travel disruptions
- Hence, scientists (social science), business leaders, security experts gather
 - Twitter data to understand human behavior and dynamics
 - For example, role of micro-influencers
 - Detection of informative tweets in crisis events

<https://nhess.copernicus.org/preprints/nhess-2020-214/nhess-2020-214.pdf>

Demographics

- Men use X more than women, accounting for almost 68% of the platform's user base
- More than 80% of Twitter's global population is under 50 years old.
 - 44% of U.S. adults aged 18-24 use Twitter.
- More teenagers are on Twitter in the U.S. than on
 - WhatsApp, Pinterest, LinkedIn, and Reddit
- Twitter ranks only slightly behind Instagram when it comes to 65+ users in the U.S. (7% vs 8%).



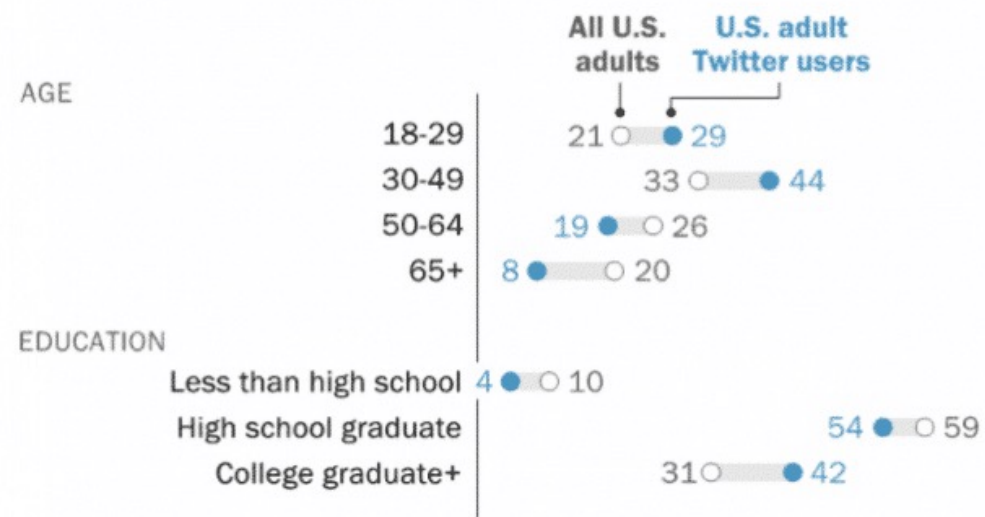
Similarly ranked sites

- 2  reddit.com
- 3  instagram.com
- 4  x.com

Income distribution – above average?

- 41% of Twitter users surveyed earn a household income about \$75,000 (vs. 32% overall)
- Only 23% of people on Twitter earn less than \$30,000 annually (vs. 30% overall)
- Education level of Twitter Users

% of _____ who are ...



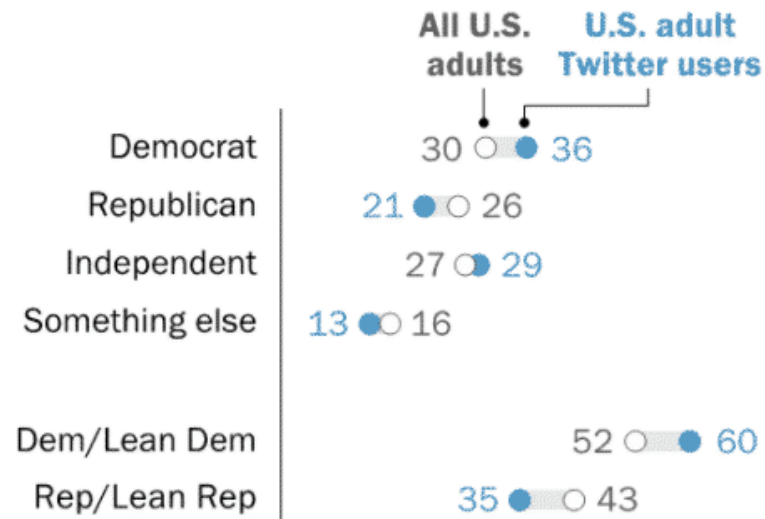
Source: *Pew Research Center*

What does this mean?
Any potential bias?

Political Beliefs

Twitter users more likely to identify as Democrats than Republicans

% of _____ who identify as...



Source: Survey of U.S. adult Twitter users conducted Nov. 21-Dec. 17, 2018, and survey of U.S. adults conducted Nov. 7-11, 2018. "Sizing Up Twitter Users"

PEW RESEARCH CENTER

What does this mean?
Any potential bias?

Bias in data

- Social media usage of the population – especially Twitter
- Lots of people use it
 - A physical sensor, like, ... make measurements
 - Social sensor, act like a physical sensor, but humans are making them
 - Examples of Twitter users sending messages even before emergency services get a call
 - A lot implicitly because they might have sent it for different purposes
 - **If we can remove noise, then we can use Tweets as social sensors**
- However, demographics are also important
 - Why?

So what did we discuss

- Social media usage of the population – especially Twitter
- Lots of people use it
 - A physical sensor, like, ... make measurements
 - Social sensor, act like a physical sensor, but humans are making them
 - Examples of Twitter users sending messages even before emergency services get a call
 - A lot implicitly because they might have sent it for different purposes
 - **If we can remove noise, then we can use Tweets as social sensors**
- **However, demographics are also important**
 - **Why?**
 - **Be careful when making inferences and generalising!**

Twitter/X Data Access

- Used to give free access to academia
- Not anymore!
- <https://developer.x.com/en/docs/x-api/getting-started/about-x-api#:~:text=Getting%20started-,About%20the%20X%20API,and%20research%20the%20public%20conversation.>

~~Firehose~~

- ~~100% data, restrictive access and costly~~

~~Gardenhose~~

- ~~Anyone to retrieve at most 1% of data~~
- ~~Sampling process is unknown (largely not specified)~~
- ~~This can be enhanced by creatively exploiting multiple accounts~~
- ~~Monthly restriction~~

~~Decahose 10%, paid and free for selected academics~~

	Free	Basic	Pro	Enterprise
Getting access	Get Started	Get Started	Get Started	Get Started
Price	Free	\$100/month	\$5000/month	
Access to X API v2	✓ (Only Post creation)	✓	✓	
Access to standard v1.1	✓ (Only Media Upload, Help, Rate Limit, and Login with X)	✓ (Only Media Upload, Help, Rate Limit, and Login with X)	✓ (Only Media Upload, Help, Rate Limit, and Login with X)	
Project limits	1 Project	1 Project	1 Project	
App limits	1 App per Project	2 Apps per Project	3 Apps per Project	
Post caps - Post	1,500	3,000	300,000	
Post caps - Pull	✗	10,000	1,000,000	
Filtered stream API	✗	✗	✓	
Access to full-archive search	✗	✗	✓	
Access to Ads API	✓	✓	✓	

It is not about Twitter but about data ...

- About unstructured and noisy data sets
- Lots of them
- What can we learn by analysing them?
- We will mainly look at Tweets and Reddit data
 - Potential usage patterns
 - Past data available or
 - are accessible!
- The techniques, approaches and methodologies we learn in this course are exploitable elsewhere



Part II

Twitter Data Structure

data is fundamental to decision making!

among all social networks twitter gives
better (*controlled!*) access

Twitter/X Home Page



Home

Explore

Notifications

Messages

Grok

Lists

Bookmarks

Jobs

Communities

Premium

Verified Orgs

Profile

More

Post

For youFollowing

1

1

posted

1.1M



Matt Lease @mattlease · Dec 16

Two Test of Time Awards! One @hcomp.conf #HCOMP2024 w/ @UTCompSci's @aasheshadri & one @ASE.conf #ASE2024 w/ @utexascece's @riponksaha, Sarfraz Khursid, & Dewayne Perry @UTGoodSystems @UTISchool @OdenInstitute @EngagingNews @UTexasResearch @AmazonScience



ischool.utexas.edu
Dr. Lease Wins Two "Test of Time" Awards in 2024
Professor Matthew Lease, who has taught at the School of Information (iSchool) since 2009, has ...

4

1

38

1.1K



Claudia Hauff    @CharlotteHase · 4h

It is this time of the year when you should think about early career researchers in the Information Retrieval field that deserve recognition! SIGIR has three categories of awards for early career researchers! sigir.org/awards/sigir-e... Self-nominations are also welcome! By Feb 16!

4

138

Pinned by people you follow



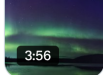
Barack Obama  @BarackObi · Dec 5



Today, I'm in Chicago for our @ObamaFoundation Democracy Forum, where we'll be bringing together leaders from around the ...



Google  @Google · Dec 13



This year, we're celebrating the B Searches of 2024 iconic performan history-making ...



Smile Train UK  @smiletrainuk Ad

This Christmas, you can give more families than ever a reason to smile.

By supporting Smile Train, you help cover the costs of comprehensive cleft treatment, transforming a child's life forever.

Where else can you make such a permanent difference in another person's life?

Search

Try Premium for free today

Upgrade your experience with less ads, power tools, and more with Premium.

Start free 14-day trial

What's happening

Sports · Trending

Joe Root

2,261 posts

Trending in United Kingdom

Chris Mason

Entertainment · Trending

Brad Pitt

3,542 posts

Music · Trending

Keith Richards

5,060 posts

[Show more](#)

Who to follow

 **Aixin Sun** 孙爱欣
@AixinSG Follows you

Follow

 **Leo Boytsov** 
@srchvrs Follows you

Follow

 **Chenyan Xiong**
@XiongChenya Follows you

Follow

[Show more](#)

[Terms of Service](#) [Privacy Policy](#) [Cookie Policy](#)

18

Data Structure (Tweet structure)

 **Developers**  
@XDevelopers · [Follow](#)

What did the developer write in their Valentine's card?

```
while(true) {  
    I = Love(You);  
}
```

7:00 PM · Feb 14, 2020

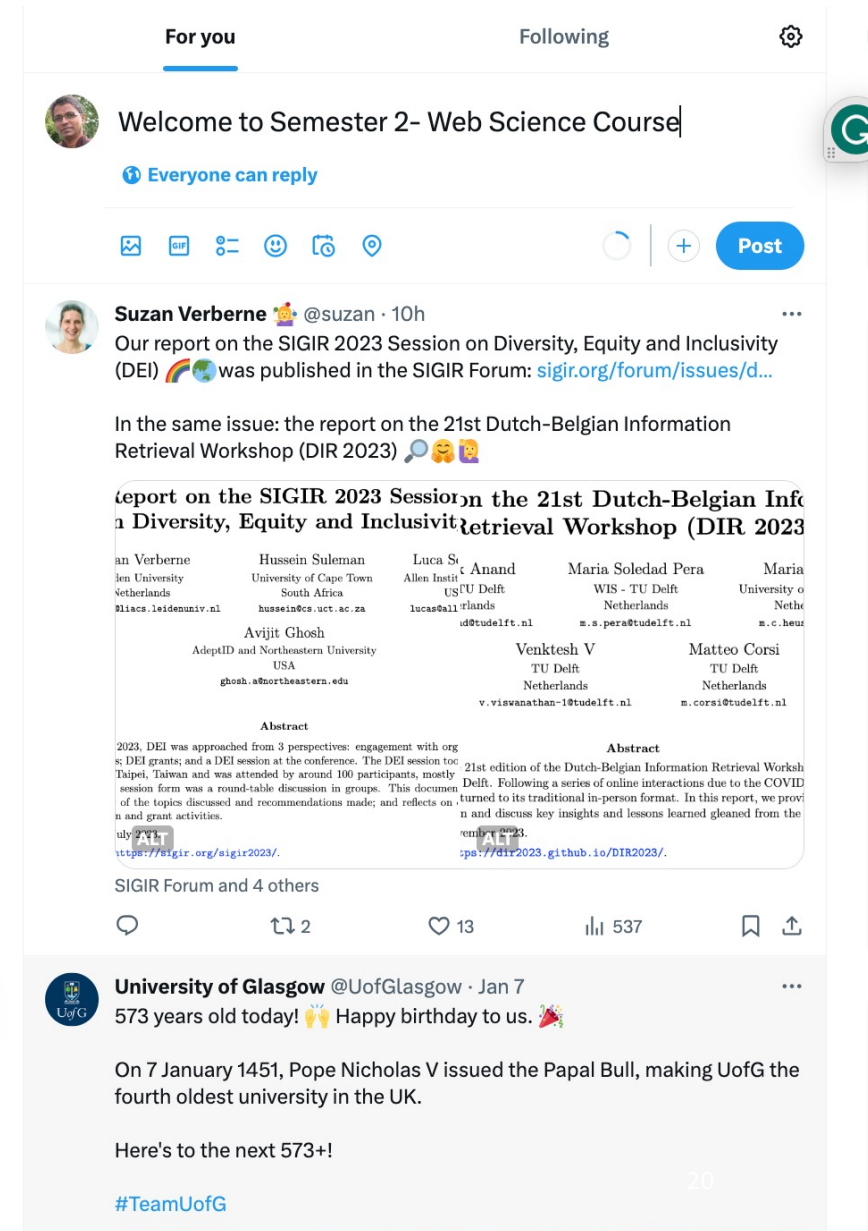
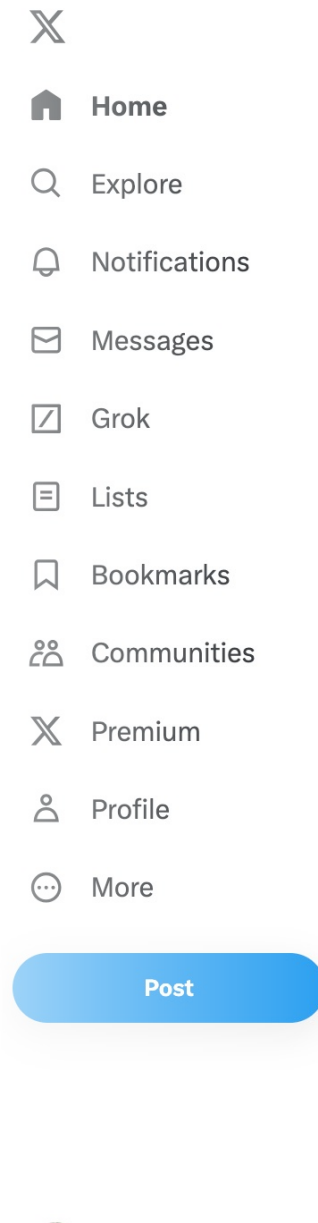
 **389**  [Reply](#)  [Copy link](#)

[Read 32 replies](#)

```
"created_at": "Fri Feb 14 19:00:55 +0000 2020",  
"id_str": "1228393702244134912",  
"text": "What did the developer write in their Valentine's card?",  
"entities": {  
  "hashtags": [],  
  "symbols": [],  
  "user_mentions": [],  
  "urls": []  
},  
"user": {  
  "entities": {  
    "url": {}  
  }  
},  
"place": {}
```

Tweet's structure

- Limited to a maximum of 280 characters (from Nov 2017)
 - It was 140 until then
 - More in certain circumstances
- Each Tweet has
 - an author, a message, a unique ID, a timestamp of when it was posted, and
 - sometimes geo metadata shared by the user.
 - Each User has a Twitter name, an ID, a number of followers, and most often an account bio.
- Tweets can be retweeted, emailed, replied to, liked, unliked and deleted.
- What can we communicate?
 - Text; URL; hashtags; mentions, picture, videos, ...
- If a Post is related to another Post, as a Retweet, reply or quote Tweet, each will be identified or embedded into the Post object.



Tweet – how does it look?



Michael Ekstrand @mdekstrand · Jan 2



Today in completely non-discoverable features: to print to PDF on an iPhone, hit Share, select Print, and then press and hold the “Print” button at the top left, regardless of what printer you have selected. (How hard would it have been to put PDF in the printer selector?)



3



Known as JSON payload →

```
{
  "created_at": "Mon Jun 28 19:56:06 +0000 2021",
  "id": 1409601521063505925,
  "id_str": "1409601521063505925",
  "text": "Good, flavoursome, aromatic beer... - Drinki",
  "truncated": false,
  "in_reply_to_status_id": null,
  "in_reply_to_status_id_str": null,
  "in_reply_to_user_id": null,
  "in_reply_to_user_id_str": null,
  "in_reply_to_screen_name": null,
  "user": {
    "id": 21012499,
    "id_str": "21012499",
    "name": "Granddad Greg",
    "screen_name": "Greg1954",
    "location": "Seaton, Devon, England",
    "url": null,
    "description": "Loves: Gilly, Tess, my Girls, Grandchildren. Likes: @TheBeachBoys, craft beer, cheese, @terryandrob, Thos Hardy, The Boss, @redsox . Dis",
    "likes": "FOOTBALL!",
    "translator_type": "none",
    "protected": false,
    "verified": false,
    "followers_count": 1205,
    "friends_count": 1458,
    "listed_count": 127,
    "favorites_count": 530,
    "statuses_count": 153966,
    "created_at": "Mon Feb 16 18:46:14 +0000 2009",
    "utc_offset": null,
    "time_zone": null,
    "geo_enabled": true,
    "lang": null,
    "contributors_enabled": false,
    "is_translator": false,
    "profile_background_color": "C0DEED",
    "profile_background_image_url": "http://abs.twimg.com/images/themes/theme1/bg.png",
    "profile_background_image_url_https": "https://abs.twimg.com/images/themes/theme1/bg.png",
    "profile_background_tile": false,
    "profile_link_color": "0084B4",
    "profile_sidebar_border_color": "FFFFFF",
    "profile_sidebar_fill_color": "DDEEF6",
    "profile_text_color": "333333",
    "profile_use_background_image": true,
    "profile_image_url": "http://pbs.twimg.com/profile_images/726701098833203201/1BP2CxrZ_normal.jpg",
    "profile_image_url_https": "https://pbs.twimg.com/profile_images/726701098833203201/1BP2CxrZ_normal.jpg",
    "profile_banner_url": "https://pbs.twimg.com/profile_banners/21012499/1367929174",
    "default_profile": false,
    "default_profile_image": false,
    "following": null,
    "follow_request_sent": null,
    "notifications": null,
    "withheld_in_countries": [],
    "geo": {
      "type": "Point",
      "coordinates": [
        [
          50.7060432,
          -3.088794
        ]
      ],
      "coordinates": {
        "type": "Point",
        "coordinates": [
          [
            -3.088794,
            50.7060432
          ]
        ],
        "place": {
          "id": "5f9dee5ce4c43c88",
          "url": "https://api.twitter.com/1.1/geo/id/5f9dee5ce4c43c88.json",
          "place_type": "city",
          "name": "Seaton",
          "full_name": "Seaton, England",
          "country_code": "GB",
          "country": "United Kingdom",
          "bounding_box": {
            "type": "Polygon",
            "coordinates": [
              [
                [
                  -3.090131,
                  50.70028
                ],
                [
                  -3.090131,
                  50.723651
                ],
                [
                  -3.060322,
                  50.723651
                ],
                [
                  -3.060322,
                  50.70028
                ]
              ]
            ]
          ],
          "attributes": {}
        }
      ],
      "contributors": null,
      "is_quote_status": false,
      "quote_count": 0,
      "reply_count": 0,
      "retweet_count": 0,
      "favorite_count": 0,
      "entities": {
        "hashtags": [],
        "urls": [
          [
            {
              "url": "https://t.co/XHiopapbEC",
              "expanded_url": "https://untappd.com/beer/s/c1045742234",
              "display_url": "untappd.com/beer/s/c1045742234",
              "indices": [
                103,
                126
              ]
            }
          ]
        ],
        "user_mentions": [],
        "symbols": []
      },
      "favorited": false,
      "retweeted": false,
      "possibly_sensitive": false,
      "filter_level": "low",
      "lang": "en",
      "timestamp_ms": "1624910166408"
    }
  }
}
```

(not for the same data....)

User Profile

Additionally

User's URL

Textual description of user and his/her interest

Verified mark if the identity of the user has been externally verified by Twitter

Profile creation date

The screenshot shows the Twitter profile of Joemon Jose. At the top, there is a header with a back arrow, the name 'Joemon Jose', and '227 Tweets'. Below this is a large banner image of a lake and mountains, and a circular profile picture of a man with glasses. To the right of the profile picture is an 'Edit profile' button. Below the banner, the name 'Joemon Jose' is followed by the handle '@joemonj'. Underneath is the bio 'IR researcher and academic', a location pin for 'Glasgow, Scotland' with a link to 'dcs.gla.ac.uk/~jj', and 'Joined March 2009'. Below this, it shows '228 Following' and '499 Followers'. At the bottom, there are four tabs: 'Tweets' (highlighted), 'Tweets & replies', 'Media', and 'Likes'.

User's real Name

User's Twitter handle @joemonj

location

User's Network activity information on Twitter

Number of tweets published by the user

Specify three significant fields in user profile and argue the reasons for their significance 22

```

{
  "created_at" : "Thu Apr 06 15:24:15 +0000 2017" ,
  "id_str" : "850006245121695744" ,
  "text" : "1\ Today we\u2019re sharing our vision for the future of the Twitter API platform!\nht" ,
  "user" : {
    "id" : 2244994945 ,
    "name" : "Twitter Dev" ,
    "screen_name" : "TwitterDev" ,
    "location" : "Internet" ,
    "url" : "https://dev.twitter.com/" ,
    "description" : "Your official source for Twitter Platform news, updates & events. Need technica"
  } ,
  "place" : {
  } ,
  "entities" : {
    "hashtags" : [
    ] ,
    "urls" : [
      {
        "url" : "https://t.co/XweGngmxlP" ,
        "unwound" : {
          "url" : "https://cards.twitter.com/cards/18ce53wgo4h/3xo1c" ,
          "title" : "Building the Future of the Twitter API Platform"
        }
      }
    ]
  } ,
  "user_mentions" : [
  ]
}

```

The user-defined location for this account's profile.
Not necessarily a location, nor machine-parsable.

Introduction to Tweet/X JSON

- Tweet Object

- [Tweet object](#)
- [User object](#), who authored the Tweet (known as user profile)
 - The User object contains Twitter User account metadata that describes the Twitter User referenced. Users can author Tweets, Retweet, quote other Users Tweets, reply to Tweets, follow Users, be @mentioned in Tweets and can be grouped into lists.
- [Entities object](#) (Arrays of hashtags, user mentions, URLs, and native media)
- [Extended entities object](#)
- [Geo object](#), – geo-information if enabled

- JSON is based on key-value pairs, with named attributes and associated values

- More details –Javascript Object Notation (JSON)

- <https://developer.twitter.com/en/docs/tweets/data-dictionary/overview/intro-to-tweet-json.html>

TweetObject

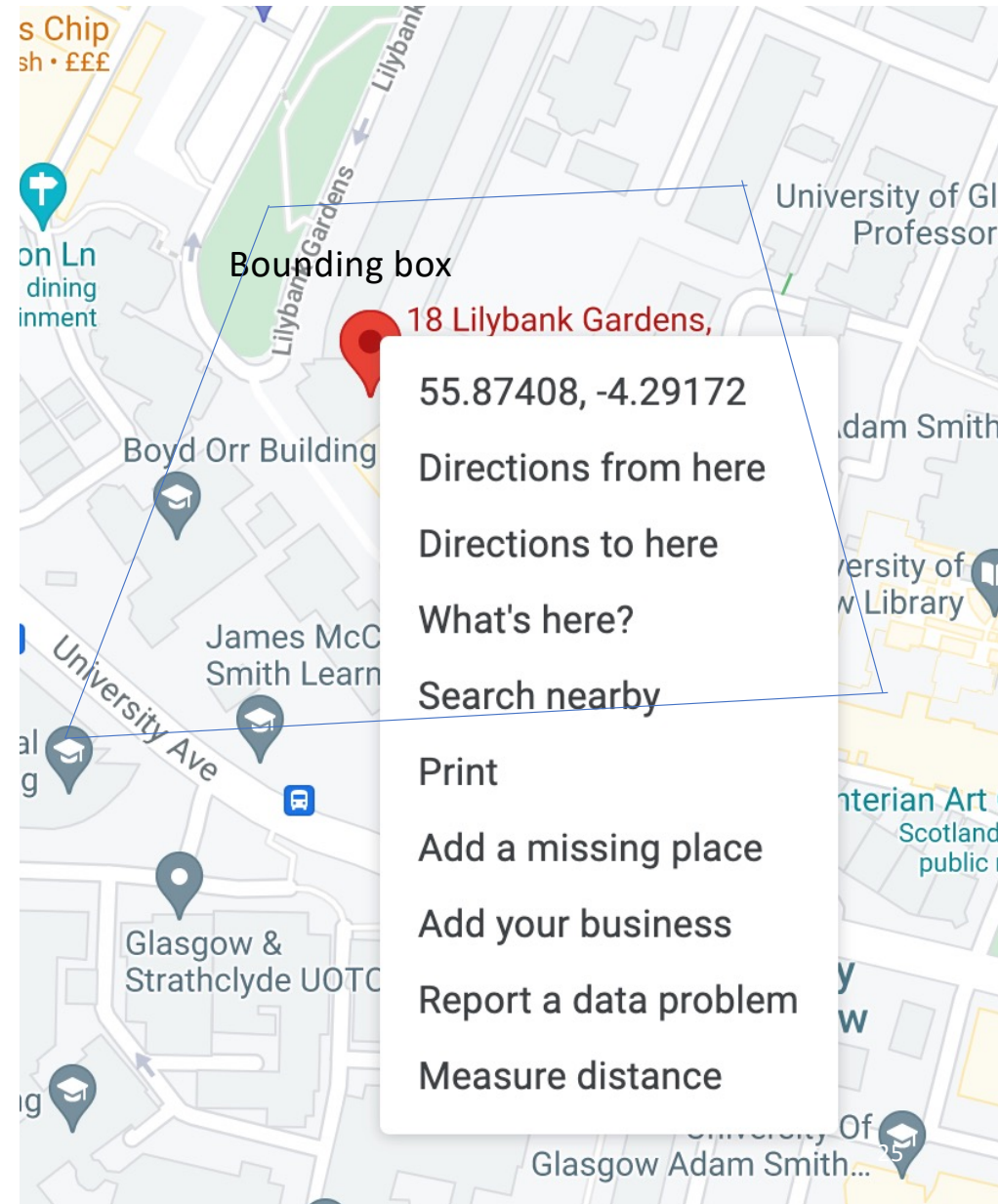


```
{
  "created_at" : "Thu May 10 15:24:15 +0000 2018" ,
  "id_str" : "850006245121695744" ,
  "text" : "Here is the Tweet message." ,
  "user" : {
  } ,
  "place" : {
  } ,
  "entities" : {
  } ,
  "extended_entities" : {
  }
}
```

Type of location information

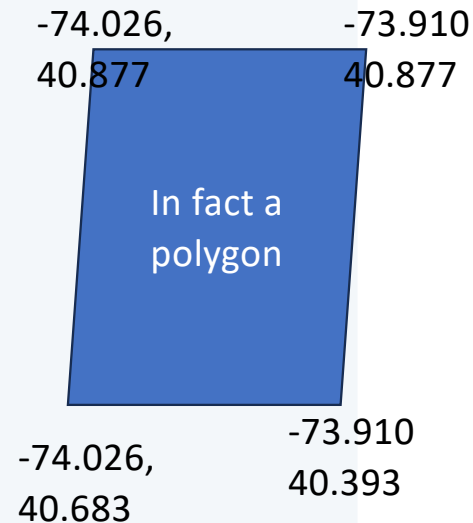
- Location information given by the user
- Geo-Location captured from the devices
- Location information inferred by Twitter software

```
{  
  "created_at": "Fri Feb 14 19:00:55 +0000 2020",  
  "id_str": "1228393702244134912",  
  "text": "What did the developer write in their Valentine's card",  
  "entities": {  
    "hashtags": [],  
    "symbols": [],  
    "user_mentions": [],  
    "urls": []  
  },  
  "user": {  
    "entities": {  
      "url": {}  
    }  
  },  
  "place": {}  
}
```



Bounding box

```
{
  "bounding_box": {
    "coordinates": [
      [
        -74.026675,
        40.683935
      ],
      [
        -74.026675,
        40.877483
      ],
      [
        -73.910408,
        40.877483
      ],
      [
        -73.910408,
        40.3935
      ]
    ],
    "type": "Polygon"
  }
}
```



- A series of longitude and latitude points,
 - defining a box which will contain the Place entity this bounding box is related to.
- Each point is an array in the form of [longitude, latitude].
- Bounding box arrays are wrapped in one additional array to be compatible with the polygon notation.

```
"coordinates": {  
  "type": "Point",  
  "coordinates": [ -73.9998279, 40.74118764 ]  
},
```

Each point is an array in the form of [longitude, latitude].

Each geo is an array in the form of [latitude, longitude].

```
{  
  "geo": {  
    "type": "Point",  
    "coordinates": [40.0160921, -105.2812196]  
  },  
  "coordinates": {  
    "type": "Point",  
    "coordinates": [-105.2812196, 40.0160921]  
  }  
}
```

Geo profile

- Tweet location is **off by default**, and you will need to **opt in** to the service.
 - "geo_enabled": true,
- Tweets with a specific latitude/longitude "geo" coordinate
- Tweets with a specific longitude/ latitude "coordinates" object
- Tweets with a Point coordinate come from GPS enabled devices and
 - represent the exact GPS location of the Tweet in question.
- This type of location does not contain any contextual information about the GPS location being referenced
 - (e.g. associated city, country, etc.),
 - unless the exact location can be associated with a Twitter Place.
- When users decide to assign a location to their Tweet, they are presented with a list of candidates
 - **Twitter Places**
 - Places are **specific, named locations with corresponding geo coordinates**.

Place object

- { "geo": null, "coordinates": null, "place": { "id": "07d9db48bc083000",
- "url": "https://api.x.com/1.1/geo/id/07d9db48bc083000.json",
- "place_type": "poi", "name": "McIntosh Lake",
- "full_name": "McIntosh Lake", "country_code": "US",
- "country": "United States",
- "bounding_box":
- { "type": "Polygon",
- "coordinates": [[[-105.14544, 40.192138], [-105.14544, 40.192138],
- [-105.14544, 40.192138], [-105.14544, 40.192138]]] },
- "attributes": { } } }

Geo Profile - Previous

Tweets with a specific latitude/longitude “geo” coordinate – deprecated
Tweets with a specific longitude/ latitude “coordinates” object

Nullable.

Represents the geographic location of this Tweet as reported by the user or client application.

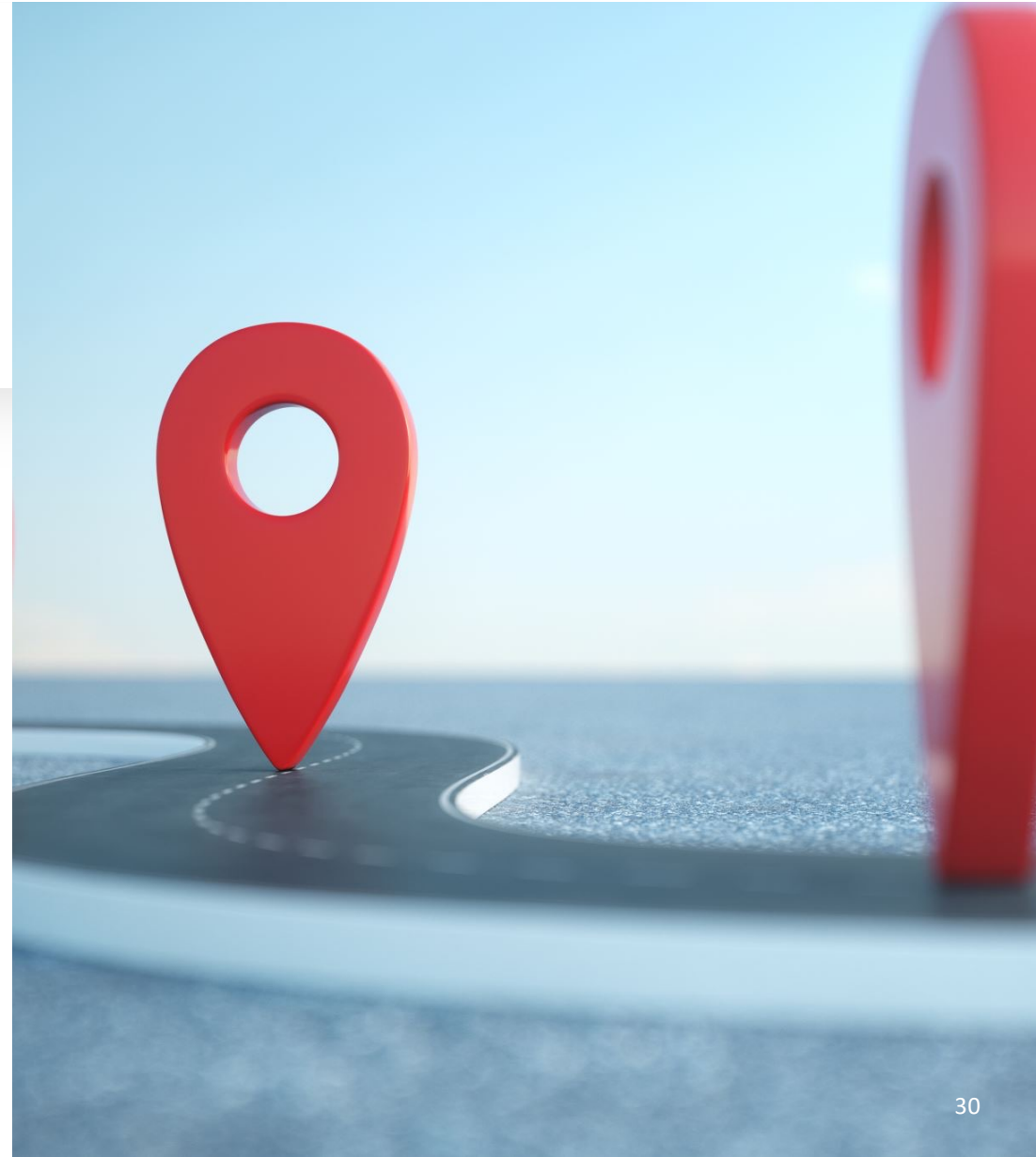
The inner coordinates array is formatted as [geoJSON](#) (longitude first, then latitude).

```
{
  "geo": {
    "type": "Point",
    "coordinates": [40.0160921, -105.2812196]
  },
  "coordinates": {
    "type": "Point",
    "coordinates": [-105.2812196, 40.0160921]
  }
}
```

“coordinates” attributes is formatted as [LONGITUDE, latitude],
~~while the “geo” attribute is formatted as [latitude, LONGITUDE].~~

Place Object

- Places are specific, named locations with corresponding geo coordinates.
- When users decide to assign a location to their Tweet, they are presented with a list of candidate Twitter Places.
- When using the API to post a Tweet, a Twitter Place can be attached by specifying a **place_id** when posting the Tweet.
- **Tweets associated with Places are not necessarily issued from that location but could also potentially be *about* that location.**



An example geo- Coordinates & Twitter Places

- `{'_id': '1409603292112502790', 'date': datetime.datetime(2021, 6, 28, 21, 3, 8), 'username': 'Natezero27', 'text': "Just posted a photo @ Fenn's, Whixall and Bettisfield Mosses National Nature Reserve https://t.co/hkBcrDoPlj", 'geo_enabled': True, 'coordinates': [-2.76446, 52.92165], 'location': None, 'place_name': 'West Midlands, England', 'place_country': 'United Kingdom', 'country_code': 'GB',`
- `'place_coordinates': [[[-3.235601, 51.825903], [-3.235601, 53.22621], [-1.172136, 53.22621], [-1.172136, 51.825903]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://instagram.com" rel="nofollow">Instagram</a>'}`

Geo-location of a tweet - summary

Geo-tagging information

- highly accurate
 - Geo_enabled = True & GPS enabled

Approximately 1-5% of tweets published on the twitter are geo-tagged

Profile of the user

"geo_enabled": false, "location": "Glasgow",

- User location can be extracted from the user field of the user profile
- Gazetter – get a location string as input and returns the **coordinates of the location** that best correspond to the string
 - Online Gazetteers – Bing, Google, MapQuest

```

85634, 'date': datetime.datetime(2021, 1, 11, 6, 11, 1), 'username': 'BritishHeraldUK', 'text':
1.50538967], 'location': 'London, United Kingdom', 'place_name': 'London, England', 'place_co
-0.187894, 51.5164655]]], 'hashtags': ['SexandtheCity'], 'mentions': [], 'source': 'dlvr.it'}
31264, 'date': datetime.datetime(2021, 1, 11, 6, 3, 32), 'username': 'BritishHeraldUK', 'text':
-0.12485951, 51.50538967], 'location': 'London, United Kingdom', 'place_name': 'London, Englar
51.5164655], [-0.187894, 51.5164655]]], 'hashtags': ['UK'], 'mentions': [], 'source': 'dlvr.it
67942, 'date': datetime.datetime(2021, 1, 11, 5, 45, 47), 'username': 'crdeliveries', 'text':
don', 'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'country_code
'mentions': [], 'source': 'Instagram')
47360, 'date': datetime.datetime(2021, 1, 11, 5, 41, 2), 'username': 'SJSChard', 'text': 'Just
'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'country_code': 'GE
ons': [], 'source': 'Instagram')
39042, 'date': datetime.datetime(2021, 1, 11, 5, 26, 1), 'username': 'BritishHeraldUK', 'text':
, 51.50538967], 'location': 'London, United Kingdom', 'place_name': 'London, England', 'place_
, [-0.187894, 51.5164655]]], 'hashtags': ['PrinceCharles'], 'mentions': [], 'source': 'dlvr.it
42592, 'date': datetime.datetime(2021, 1, 11, 5, 21, 15), 'username': 'trafficgifs', 'text': '
Y', 'place_name': 'Lambeth, London', 'place_country': 'United Kingdom', 'country_code': 'GB',
': [], 'source': 'everybot'}
73891, 'date': datetime.datetime(2021, 1, 11, 5, 20, 6), 'username': 'PlazaEstates', 'text': '
inates': [-0.164204, 51.515447], 'location': 'UK', 'place_name': 'Paddington, London', 'place_
7916], [-0.2160701, 51.5397916]]], 'hashtags': [], 'mentions': [], 'source': 'Estate Agent Fee
09121, 'date': datetime.datetime(2021, 1, 11, 5, 15, 24), 'username': 'WhatsOnOLIO', 'text': '
n': '', 'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'country_cc
London', 'UnitedKingdom', 'foodwaste', 'free'], 'mentions': [], 'source': 'OLIO Articles Plugi
37152, 'date': datetime.datetime(2021, 1, 11, 5, 15, 15), 'username': 'WhatsOnOLIO', 'text': '
cation': '', 'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'count
': ['London', 'UnitedKingdom', 'foodwaste', 'free'], 'mentions': [], 'source': 'OLIO Articles
50145, 'date': datetime.datetime(2021, 1, 11, 5, 13, 15), 'username': 'WhatsOnOLIO', 'text': '
cation': '', 'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'count
': ['London', 'UnitedKingdom', 'foodwaste', 'free'], 'mentions': [], 'source': 'OLIO Articles
08736, 'date': datetime.datetime(2021, 1, 11, 5, 12, 32), 'username': 'WhatsOnOLIO', 'text': '
1], 'location': '', 'place_name': 'City of London, London', 'place_country': 'United Kingdom',
ashtags': ['London', 'UnitedKingdom', 'foodwaste', 'free'], 'mentions': [], 'source': 'OLIO Ar
58115, 'date': datetime.datetime(2021, 1, 11, 5, 4, 11), 'username': 'GMSdotnet', 'text': 'Jus
'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'country_code': 'C
ions': [], 'source': 'Instagram')
90208, 'date': datetime.datetime(2021, 1, 11, 4, 44, 41), 'username': 'prince darien', 'text':
don, England, UK', 'place_name': 'City of London, London', 'place_country': 'United Kingdom',
shtags': [], 'mentions': [], 'source': 'Instagram')
79332, 'date': datetime.datetime(2021, 1, 11, 4, 33, 54), 'username': 'WhatsOnOLIO', 'text': '
ation': '', 'place_name': 'Camberwell, London', 'place_country': 'United Kingdom', 'country_cc
ags': ['London', 'UnitedKingdom', 'foodwaste', 'free'], 'mentions': [], 'source': 'OLIO Articl
26178, 'date': datetime.datetime(2021, 1, 11, 4, 32, 45), 'username': 'GMSdotnet', 'text': 'Jl
'place_name': 'City of London, London', 'place_country': 'United Kingdom', 'country_code': '
tions': [], 'source': 'Instagram')
35043, 'date': datetime.datetime(2021, 1, 11, 4, 31, 52), 'username': 'BBang7891', 'text': '@c
', 'place_name': 'Islington, London', 'place_country': 'United Kingdom', 'country_code': 'GB'
mentions': ['c', 'Complex'], 'source': 'Instagram')
00096, 'date': datetime.datetime(2021, 1, 11, 4, 5, 34), 'username': 'dapperlytamed', 'text':
'coordinates': [-0.1758279, 51.49624537], 'location': 'India', 'place_name': 'Kensington, Lor
97908, 51.5303477], [-0.2285894, 51.5303477]]], 'hashtags': [], 'mentions': [], 'source': 'Ins
70625, 'date': datetime.datetime(2021, 1, 11, 3, 53, 33), 'username': '24hrRock', 'text': 'Ste
[-0.10739665, 51.50538977], 'location': 'UK', 'place_name': 'Lambeth, London', 'place_country'
1, 51.509887]]], 'hashtags': [], 'mentions': [], 'source': 'dlvr.it'}
97316, 'date': datetime.datetime(2021, 1, 11, 3, 31, 53), 'username': 'shehshukla11', 'text':
tionalspeaker #mondaymorning #mondayblues #vibes #morningvibes https://t.co/8Jc1C3u94j', 'geoe
Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-0.112442, 51.5068], [-0.0733794, 51.
yvibes', 'practice', 'soul', 'motivational', 'motivationalspeaker', 'mondaymorning', 'mondaybl
23042, 'date': datetime.datetime(2021, 1, 11, 3, 20, 11), 'username': 'jerzeej111', 'text': 'C
t Earth', 'place_name': 'Lambeth, London', 'place_country': 'United Kingdom', 'country_code':
ntions': [], 'source': 'Instagram')
nt call last):

```

INSTAGRAM

Place coordinates

```
{ 'id': '1409603773274726404', 'date': datetime.datetime(2021, 6, 28, 21, 5, 3), 'username': 'ego_leeds', 'text': 'Just posted a photo @ Leeds, England, UK http://t.co/CkH5yHonPM', 'geoenabled': True, 'coordinates': [-1.52145834, 53.81177433], 'location': 'Leeds uk', 'place_name': 'Leeds, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-1.706057, 53.736369], [-1.706057, 53.867543], [-1.424923, 53.867543], [-1.424923, 53.736369]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://instagram.com" rel="nofollow">Instagram</a>'}
{ 'id': '1409603773413179398', 'date': datetime.datetime(2021, 6, 28, 21, 5, 3), 'username': 'danniellebosxx', 'text': 'Just cried at the sound of love island ahaha my hormones', 'geoenabled': True, 'coordinates': None, 'location': None, 'place_name': 'Castleford, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-1.400752, 53.708192], [-1.400752, 53.738875], [-1.298097, 53.738875], [-1.298097, 53.708192]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '140960377562252808', 'date': datetime.datetime(2021, 6, 28, 21, 5, 3), 'username': 'kyriejs_', 'text': 'My MD said I can leave at half 4 tomorrow to make sure I get home for the game what a guy', 'geoenabled': True, 'coordinates': None, 'location': 'Birmingham, England', 'place_name': 'Birmingham, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-2.033651, 52.381063], [-2.033651, 52.60687], [-1.74763, 52.60687], [-1.74763, 52.381063]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603776940589057', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'ecm_242', 'text': 'If you need a man to open a bottle of Prosecco, you're doing it wrong. 🍷👉👉👉 #LoveIsland', 'geoenabled': True, 'coordinates': None, 'location': None, 'place_name': 'Crosslee, Scotland', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-4.585836, 55.85653], [-4.585836, 55.872931], [-4.52392, 55.872931], [-4.52392, 55.85653]]], 'hashtags': ['LoveIsland'], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603777489952776', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'ClareS_J', 'text': 'This is genuinely 🥰', 'geoenabled': True, 'coordinates': None, 'location': 'Edinburgh', 'place_name': 'Scotland, United Kingdom', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-7.660857, 54.633631], [-7.660857, 54.633631], [-0.740825, 54.633631]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '140960377765720834', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'andrew14ad', 'text': '@RedRoseMichelle Nice to see them struggle as they have been half arsing it all tournament', 'geoenabled': True, 'coordinates': None, 'location': 'Gateshead, Tyne and Wear', 'place_name': 'Gateshead, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-1.619675, 54.91311], [-1.619675, 54.96998], [-1.517516, 54.96998], [-1.517516, 54.91311]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/#/download/ipad" rel="nofollow">Twitter for iPad</a>'}
{ 'id': '1409603778026868736', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'driverbod125', 'text': 'This looks fun!', 'geoenabled': True, 'coordinates': None, 'location': 'Bristol', 'place_name': 'South West, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-3.68504, 49.882472], [-3.68504, 52.112542], [-1.485734, 52.112542], [-1.485734, 49.882472]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603778874167304', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'monnie_mons', 'text': 'Wow she looks stunning in yellow 🧡', 'geoenabled': True, 'coordinates': None, 'location': 'London, England', 'place_name': 'London, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-0.510365, 51.286702], [-0.510365, 51.691824], [0.334043, 51.691824], [0.334043, 51.286702]]], 'hashtags': [], 'mentions': [], 'source': '<a href="http://twitter.com/download/android" rel="nofollow">Twitter for Android</a>'}
{ 'id': '1409603779879688193', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'Roger_Xanth_Day', 'text': '@Freddiesharp12 @jackalsbynight @APHClarkson How slow it is we sink', 'geoenabled': True, 'coordinates': None, 'location': 'Bristol', 'place_name': 'Bristol, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-2.659936, 51.399367], [-2.659936, 51.516387], [-2.510844, 51.516387], [-2.510844, 51.399367]]], 'hashtags': [], 'mentions': ['Freddiesharp12', 'jackalsbynight', 'APHClarkson'], 'source': '<a href="http://twitter.com/download/android" rel="nofollow">Twitter for Android</a>'}
{ 'id': '1409603779373240328', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'lochdara', 'text': '@RoisinTCDNeuro It was a pleasure .', 'geoenabled': True, 'coordinates': None, 'location': 'Wexford, Ireland', 'place_name': 'Wexford, Ireland', 'place_country': 'Ireland', 'country_code': 'IE', 'place_coordinates': [[[-7.017507, 52.122381], [-7.017507, 52.797086], [-6.141269, 52.797086], [-6.141269, 52.122381]]], 'hashtags': [], 'mentions': ['RoisinTCDNeuro'], 'source': '<a href="http://twitter.com/download/android" rel="nofollow">Twitter for Android</a>'}
{ 'id': '1409603779767504896', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'debsknig', 'text': '@TBanton18 is playing a lovely innings this evening, so good to see him going well 57 from 28 so far #T20Blast2021', 'geoenabled': True, 'coordinates': None, 'location': 'Essex', 'place_name': 'Chelmsford, East', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[0.435659, 51.698636], [0.435659, 51.78115], [0.521299, 51.78115], [0.521299, 51.698636]]], 'hashtags': ['T20Blast2021'], 'mentions': ['TBanton18'], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603781759848450', 'date': datetime.datetime(2021, 6, 28, 21, 5, 5), 'username': 'andrew_jeans', 'text': "@PipSinks @shiv5468 @history99917180 @craigymcmillan Was a 85 plate 2x2..( if the passengers in the back were 2'6 max🤔), but indeed, lots of fun...", 'geoenabled': True, 'coordinates': None, 'location': 'Scotland, United Kingdom (tbc)', 'place_name': 'Scotland, United Kingdom', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-7.660857, 54.633631], [-7.660857, 54.633631], [-0.740825, 54.633631], [-0.740825, 54.633631]]], 'hashtags': [], 'mentions': ['PipSinks', 'shiv5468', 'history99917180', 'craigymcmillan'], 'source': '<a href="http://twitter.com/download/android" rel="nofollow">Twitter for Android</a>'}
{ 'id': '1409603783143927816', 'date': datetime.datetime(2021, 6, 28, 21, 5, 5), 'username': 'JulieWard_', 'text': '@mairtinofaolain @gillian @KevinOMS364116: Always 🥰', 'geoenabled': True, 'coordinates': None, 'location': 'Waterford, Ireland', 'place_name': 'Waterford, Ireland', 'place_country': 'Ireland', 'country_code': 'IE', 'place_coordinates': [[[-8.162198, 51.937671], [-8.162198, 52.364327], [-6.950104, 52.364327], [-6.950104, 51.937671]]], 'hashtags': [], 'mentions': ['mairtinofaolain', 'o_gillian', 'KevinOMS364116'], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603785140363266', 'date': datetime.datetime(2021, 6, 28, 21, 5, 6), 'username': 'kate_snow', 'text': 'Maaaaaan I already really want to go on a holiday abroad #loveisland2021', 'geoenabled': True, 'coordinates': None, 'location': 'Eastleigh, Hampshire', 'place_name': 'Eastleigh, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-1.413732, 50.942936], [-1.413732, 51.004389], [-1.334973, 51.004389], [-1.334973, 50.942936]]], 'hashtags': ['loveisland2021'], 'mentions': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603785714974730', 'date': datetime.datetime(2021, 6, 28, 21, 5, 6), 'username': 'FernDavies', 'text': 'My first induction has got off to a great start! A pleasure to greet 30 new RLC soldiers yesterday and guide them in their first week in the regiment. It's going to be fun! Exciting working with the future of the Corps! @UKArmyLogistics @25RLC @BritishArmy 🇬🇧', 'geoenabled': True, 'coordinates': None, 'location': 'North Wales/Beverley', 'place_name': 'Leconfield, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-0.467151, 53.869332], [-0.467151, 53.888463], [-0.422683, 53.888463], [-0.422683, 53.869332]]], 'hashtags': [], 'source': '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>'}
{ 'id': '1409603785979215875', 'date': datetime.datetime(2021, 6, 28, 21, 5, 6), 'username': 'AllanThornley1', 'text': '@IpswichTown Great signing. Vavlay is a smashing goalie', 'geoenabled': True, 'coordinates': None, 'location': 'Glasgow, Scotland', 'place_name': 'Glasgow, Scotland', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-4.393285, 55.796184], [-4.393285, 55.920421], [-4.090218, 55.920421], [-4.090218, 55.796184]]], 'hashtags': [], 'mentions': ['IpswichTown'], 'source': '<a href="http://twitter.com/download/android" rel="nofollow">Twitter for Android</a>'}
@Teebook (most recent 311 likes):
```

Entity Objects

```
"entities": {  
  "hashtags": [ ],  
  "urls": [ ],  
  "user_mentions": [ ],  
  "symbols": [ ]  
}
```

- each Tweet have "entity" objects,
 - which are arrays of common Tweet contents
 - such as hashtags, mentions, media, and links.
 - If there are links, the JSON payload can also provide metadata such as the fully unwound URL and the webpage's title and description.

#nodejs

```
{ "hashtags": [ { "indices": [ 32, 38 ], "text": "nodejs" } ] }
```

```

{
  "created_at" : "Thu May 10 17:41:57 +0000 2018" ,
  "id_str" : "994633657141813248" ,
  "text" : "Just another Extended Tweet with more than 140 characters, generated as a documentati
  "display_text_range" : [ 0 , 140 ] ,
  "truncated" : true ,
  "user" : {
    "id_str" : "944480690" ,
    "screen_name" : "FloodSocial"
  } ,
  "extended_tweet" : {
    "full_text" : "Just another Extended Tweet with more than 140 characters, generated as a doc
    "display_text_range" : [ 0 , 249 ] ,
    "entities" : {
      "hashtags" : [ {
        "text" : "documentation" ,
        "indices" : [ 211 , 225 ]
      } , {
        "text" : "parsingJSON" ,
        "indices" : [ 226 , 238 ]
      } , {
        "text" : "GeoTagged" ,
        "indices" : [ 239 , 249 ]
      } ]
    }
  } ,
  "entities" : {
    "hashtags" : [ ]
  }
}

```

27 Nov 2023 —

Twitter API v2 is built from the ground up to support **Tweets** with up to 280 characters. With v2, there is no concept of **'extended' Tweets**.

X extended entities

- The `extended_entities` object contains
- a single `media` array of `media` objects (see the `entities` section for its data dictionary).
- No other entity types, such as hashtags and links, are included in the `extended_entities` section.
- The `media` object in the `extended_entities` section is identical in structure to the one included in the `entities` section.



API demos
@FloodSocial



Test Tweet with [@mentionThis](#) [\\$twtr](#) [bit.ly/2pUk4be](#) [#hashtag](#)



6:02 PM · May 8, 2017 from Longmont, CO



Post your reply

Reply

[Show probable spam](#)

Tutorial Questions?

- Why do you think location information is important?
 - Hint - Think about potential applications
- What are the benefits of location information?
 - Hint – what kind of inferences we could make if we have location information
- How does Twitter encode location information & their advantages and limitations?
 - Hint - <https://developer.twitter.com/en/docs/twitter-api/v1/data-dictionary/object-model/geo>